

Multiple Choice (10 marks)

1. Consider the Bayesian network given below. How many independent parameters are needed for this Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$?
 - (a) 2
 - (b) 4
 - (c) 8
 - (d) 16
2. Statement 1| RELUs are not monotonic, but sigmoids are monotonic. Statement 2| Neural networks trained with gradient descent with high probability converge to the global optimum.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Suppose your model is overfitting. Which of the following is NOT a valid way to try and reduce the overfitting?
 - (a) Increase the amount of training data.
 - (b) Improve the optimisation algorithm being used for error minimisation.
 - (c) Decrease the model complexity.
 - (d) Reduce the noise in the training data.
4. Which of the following best describes the joint probability distribution $P(X, Y, Z)$ for the given Bayes net. $X \leftarrow Y \rightarrow Z$?
 - (a) $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$
 - (b) $P(X, Y, Z) = P(X) * P(Y|X) * P(Z|Y)$
 - (c) $P(X, Y, Z) = P(Z) * P(X|Z) * P(Y|Z)$
 - (d) $P(X, Y, Z) = P(X) * P(Y) * P(Z)$
5. Which of the following best describes what discriminative approaches try to model? (w are the parameters in the model)
 - (a) $p(y|x, w)$
 - (b) $p(y, x)$
 - (c) $p(w|x, w)$
 - (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: Overfitting happens when a model is too simple and cannot accurately capture the training data or generalize to new data.
7. True or False: In Naive Bayes, attributes are statistically independent of one another given the class value.
8. True or False: The variance of the Maximum A Posteriori (MAP) estimate is lower than that of the Maximum Likelihood Estimate (MLE).
9. True or False: If $P(A, B)$ decreases while $P(A)$ increases, then $P(B|A)$ decreases in accordance with conditional probability principles.
10. True or False: The cost of one gradient descent update, given the gradient, is $O(D)$, where D represents the number of dimensions in the feature space.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the assumption of full versus diagonal class covariance matrices in a Gaussian Bayes classifier influences the balance between underfitting and overfitting.
12. Discuss how to generate a 10×5 Gaussian matrix with entries sampled from $\mathcal{N}(\mu = 5, \sigma^2 = 16)$ and a 10×10 uniform matrix from $U[-1, 1)$. Include the relevant PyTorch commands and the rationale behind your choices.

End of Paper